

Clinical Evidence Review

Case BV-151

Generated 2026-10-07 14:33:45

Study Chest X-ray · 1222×1097

Current presentation

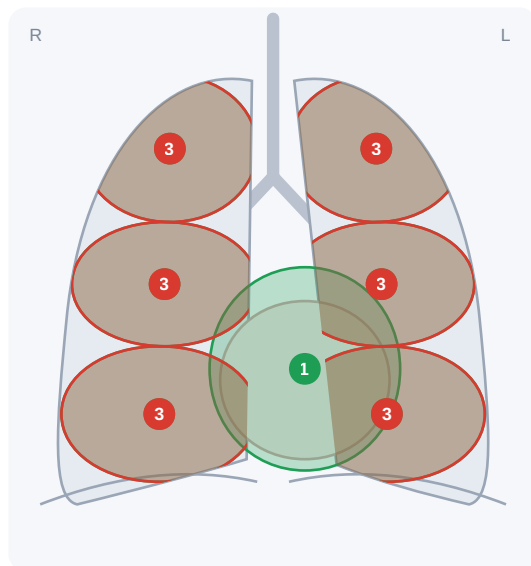
Worsening shortness of breath for 3 days, can't lie flat, waking up breathless at night, ankle swelling. No fever, no cough.

Present: Shortness of breath (3 days), Orthopnea, Night-time breathlessness, Ankle / leg swelling **Denied:** Fever, Cough

Relevant patient history

2026-08-12	Pulmonary edema	Radiology report · case_a_history.pdf · page 3
2026-08-12	Cardiomegaly	Radiology report · case_a_history.pdf · page 3
2026-07-30	Congestive heart failure	Discharge summary · case_a_history.pdf · page 1
2026-07-30	Diuretic therapy	Discharge summary · case_a_history.pdf · page 1
2026-07-28	Cardiomyopathy	Echo report · case_a_history.pdf · page 2
2026-07-28	Valvular heart disease	Echo report · case_a_history.pdf · page 2
2012	Hypertension	Discharge summary · case_a_history.pdf · page 1

Where the findings are



1 Cardiomegaly — supported

cardiac silhouette · model outline

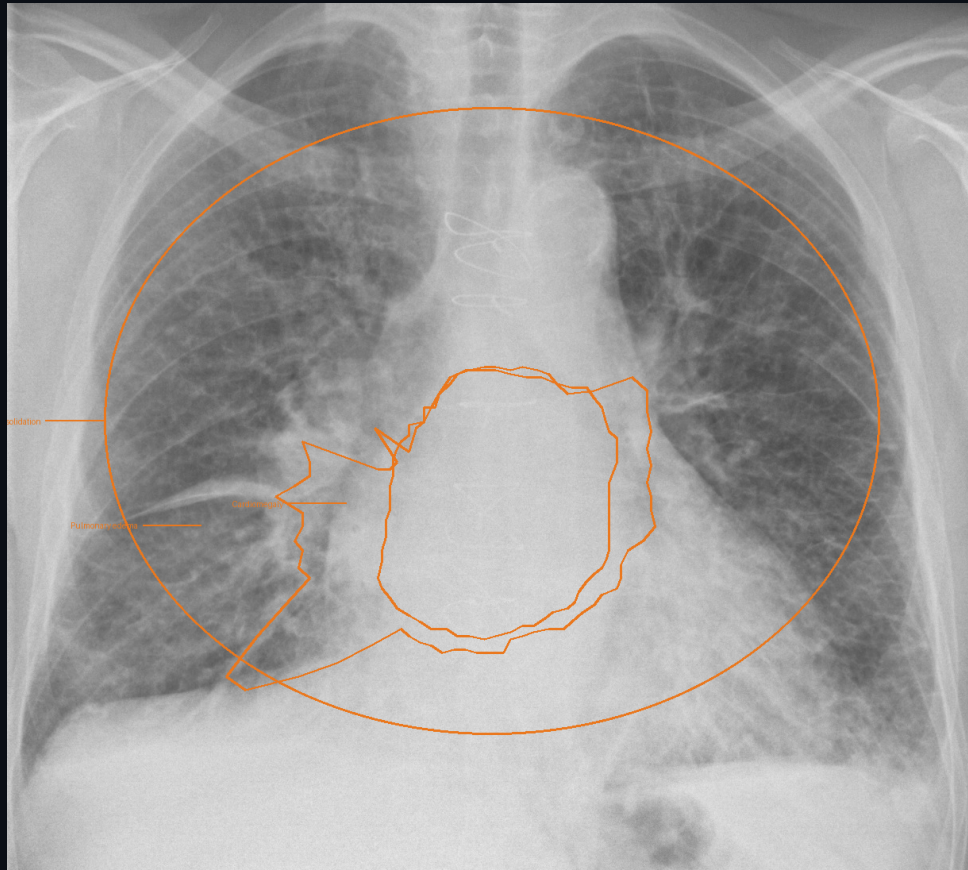
2 Pulmonary edema — supported

right upper zone, right mid zone, right lower zone, left upper zone, left mid zone, left lower zone · model outline

3 Consolidation — conflicting

right upper zone, right mid zone, right lower zone, left upper zone, left mid zone, left lower zone · approximate zone

Chest X-ray assessment



3 candidate finding(s): 2 supported, 1 conflicting. Overall evidence quality: **MODERATE**.

Finding 01 **Cardiomegaly**

Supported

Consider cardiomegaly (Heart). Imaging and the supplied context agree.

IMAGE EVIDENCE

- The heart appears enlarged
- Primary image signal: strong
- Independent verifier: strong
- “chronic recurrent congestive heart failure”
- “background changes of mild congestive heart failure”
- Localized to Heart · segmented

RELEVANT HISTORY

- Cardiomegaly · 2026-08-12
Radiology report · case_a_history.pdf · page 3
- Congestive heart failure · 2026-07-30
Discharge summary · case_a_history.pdf · page 1
- Cardiomyopathy · 2026-07-28
Echo report · case_a_history.pdf · page 2
- Valvular heart disease · 2026-07-28
Echo report · case_a_history.pdf · page 2
- Hypertension · 2012
Discharge summary · case_a_history.pdf · page 1

CURRENT PRESENTATION

- Shortness of breath · 3 days
- Orthopnea
- Ankle / leg swelling

CONFLICTING / NEGATIVE EVIDENCE

None identified

EVIDENCE STRENGTH

HIGH

Image confidence 90 % · CLEAR 94 %,
CheXzero 87 %
calibrated on 202 labelled films (CheXpert)

Finding 02 **Pulmonary edema**

Supported

Consider pulmonary edema (Lung). Imaging and the supplied context agree.

IMAGE EVIDENCE

- Visual reasoning model did not identify this finding
- Primary image signal: strong
- Independent verifier: strong
- “chronic recurrent pulmonary edema”
- “pulmonary edema superimposed on interstitial changes”
- Localized to Lung · segmented

RELEVANT HISTORY

- Pulmonary edema · 2026-08-12
Radiology report · case_a_history.pdf · page 3
- Congestive heart failure · 2026-07-30
Discharge summary · case_a_history.pdf · page 1
- Diuretic therapy · 2026-07-30
Discharge summary · case_a_history.pdf · page 1
- Cardiomyopathy · 2026-07-28
Echo report · case_a_history.pdf · page 2
- Valvular heart disease · 2026-07-28
Echo report · case_a_history.pdf · page 2

CURRENT PRESENTATION

- Shortness of breath · 3 days
- Orthopnea
- Night-time breathlessness
- Ankle / leg swelling

CONFLICTING / NEGATIVE EVIDENCE

None identified

EVIDENCE STRENGTH

HIGH

Image confidence 88 % · CLEAR 79 %,
CheXzero 97 %
calibrated on 202 labelled films (CheXpert)

Finding 03 Consolidation

Conflicting

Evidence for consolidation is inconsistent. Human review required.

IMAGE EVIDENCE

- Visual reasoning model did not identify this finding
- Primary image signal: moderate
- Independent verifier: strong
- “pulmonary edema or pneumonia superimposed on chronic interstitial lung disease”
- “dependent edema or superimposed pneumonia”

RELEVANT HISTORY

None identified

CURRENT PRESENTATION

- Shortness of breath · 3 days

CONFLICTING / NEGATIVE EVIDENCE

- Image signal present, but the current presentation argues against it: fever denied, cough denied.
- Fever denied
- Cough denied

EVIDENCE STRENGTH

MODERATE

Image confidence 31 % · CLEAR 28 %,
CheXzero 34 %
calibrated on 202 labelled films (CheXpert)

Other observations (single reader, unverified)

- Interstitial thickening — lung fields MedGemma 1.5 · CLEAR agrees

Since last report (2026-08-12)

- **NOT SEEN NOW** — Pleural effusion. Prior: “Small right-sided pleural effusion.” case_a_history.pdf p.3
- **KNOWN** — Cardiomegaly. Prior: “Impression: Cardiomegaly with features of pulmonary venous congestion.” case_a_history.pdf p.3
- **KNOWN** — Pulmonary edema. Prior: “Upper lobe venous diversion.” case_a_history.pdf p.3

Checked and rejected

- **Sternal wires** MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.81 < 0.85)
- **Right pleural effusion** MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.37 < 0.85)
- **Left pleural effusion** MedGemma 1.5 — CLEAR, reading the same phrase against the film, disagrees (0.73 < 0.85)
- **Pleural effusion** CLEAR + CheXzero — prompt score was high, but no phrase among the film's top 300 of 368,294 report phrases names it (best #354)
- **Atelectasis** CLEAR + CheXzero — prompt score was high, but no phrase among the film's top 300 of 368,294 report phrases names it (best #633)
- **Pulmonary fibrosis** CLEAR concepts — the film's closest report phrases suggest it, but MedGemma did not confirm it on the image
- **Lines & devices** CLEAR + CheXzero — prompt score was high, but no phrase among the film's top 300 of 368,294 report phrases names it (best #612)
- **Pneumonia** label hierarchy — already explained by consolidation
- **Widened mediastinum** label hierarchy — already explained by cardiomegaly

Limitations

Findings are candidate observations generated by automated models and reconciled against the supplied documents. Evidence strength is a rule-based summary of agreement between sources. Image confidence is each image reader's probability calibrated (Platt scaling) on labelled films, and reflects the image only; the prevalence in clinical use may differ from the calibration set. Localization is approximate. Models used: CLEAR (primary), CheXzero (verifier), MedGemma 1.5 4B (localization and visual description).

Review statement

This report is generated as clinical decision support from the supplied imaging and contextual evidence. It is not a definitive diagnosis and requires review by a qualified clinician.